

Course One

Foundations of Data Science



Instructions

Use this PACE strategy document to record decisions and reflections as you work through this end-of-course project. You can use this document as a guide to consider your responses and reflections at different stages of the data analytical process. Additionally, the PACE strategy documents can be used as a resource when working on future projects.

Course Project Recap

Regardless of which track you have chosen to complete, your goals for this project are:

- ☐ Complete the PACE Strategy Document to plan your project while considering your audience members, teammates, key milestones, and overall project goal.
- ☐ Create a project proposal for the data team.

Relevant Interview Questions

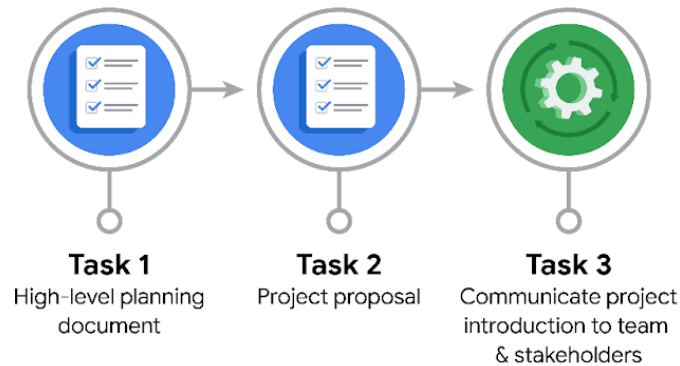
Completing this end-of-course project will empower you to respond to the following interview topics:

- As a new member of a data analytics team, what steps could you take to get 'up to speed' with a current project? What steps would you take? Who would you like to meet with?
- How would you plan an analytics project?
- What steps would you take to translate a business question to an analytical solution?
- Why is actively managing data an important part of a data analytics team's responsibilities?
- What are some considerations you might need to be mindful of when reporting results?



Reference Guide

This project has three tasks; the following visual identifies how the stages of PACE are incorporated across those tasks.



Data Project Questions & Considerations



PACE: Plan Stage

- Who is your audience for this project?

New York City Taxi and Limousine Commission (TLC)

- What are you trying to solve or accomplish? And, what do you anticipate the impact of this work will be on the larger needs of the client?

Estimate the taxi fare before the ride, based on identified variables. Positively impacting 200k taxi drivers.

- What questions need to be asked or answered?

What is the condition of the data?

What variables can be most useful in this project?

What trends in the data can provide valuable insights?

What steps can be taken to reduce the impact of bias?

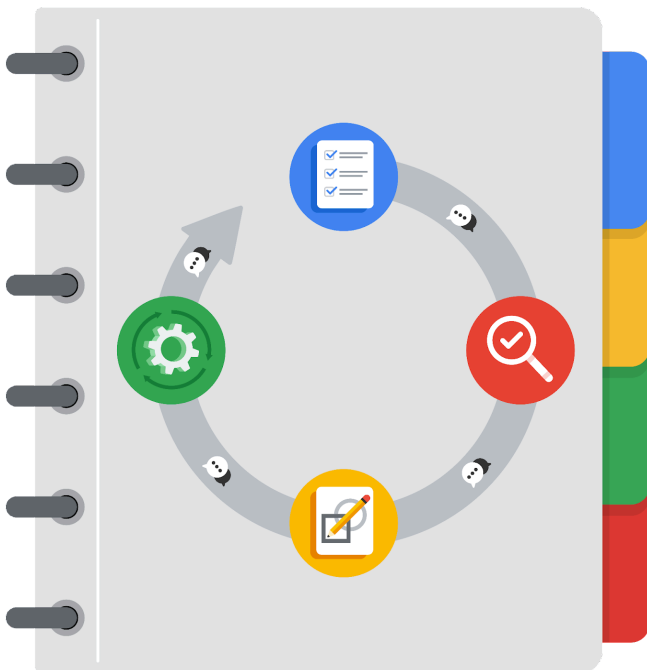
- What resources are required to complete this project?

Dataset, Python notebook and stakeholder and stakeholders input.

- What are the deliverables that will need to be created over the course of this project?

The deliverables include data cleaning for exploratory data analysis, visualisations, statistics models, regressions analysis and/or machine learning models.

THE PACE WORKFLOW



[Alt-text: The PACE Workflow with the four stages in a circle: plan, analyze, construct, and execute.]

You have been asked to demonstrate for the company's data team how you would use the PACE workflow to organize and classify tasks for the upcoming project. Select a PACE stage from the dropdown buttons. A few tasks involve more than one stage of the PACE workflow. Additionally, not every workplace scenario will require every task. Refer back to the Course 1 end-of-course portfolio project overview reading if you need more information about the tasks within the project.



Project tasks

Following are a group of tasks your company's data team has determined need to be completed within this project. The data analysis manager has asked you to organize these tasks in preparation for the project proposal document. First, identify which stage of the PACE workflow each task would best fit under using the drop down menu. Next, give an explanation of why you selected the stage for each task. Review the following readings to help guide your selections and explanation: The PACE stages and Communicate objectives with a project proposal. You will later reorder these tasks within a project proposal.

1. Evaluating the model: **Execute** ▾

Why did you select this stage for this task?

After the model is complete the data will be run through to ensure that the result meets the project expectations and results.

2. Conduct hypothesis testing: **Analyze** ▾ and **Construct** ▾

Why did you select these stages for this task?

During the analysis phase, it is determined that a statistical test will be used. During the construction phase, the test is carried out.

3. Begin exploring the data: **Analyze** ▾

Why did you select this stage for this task?

During the analysis Phase I will explore the data set and get familiar with the trends within the data.

4. Data exploration and cleaning: **Plan** ▾ and **Analyze** ▾

Why did you select these stages for this task?



During the planning phase I will determine what relevant variables to look for. During the analyze phase I will clean the data.

5. **Establish structure for project workflow (PACE):** Plan ▾

Why did you select this stage for this task?

Create the initial PACE workflow and outline the best approach for the project.

6. **Communicate final insights with stakeholders:** Execute ▾

Why did you select this stage for this task?

Communicating the final insights to stakeholders is part of the final stages of the projects. Although communication is necessary throughout the project.

7. **Compute descriptive statistics:** Analyze ▾

Why did you select this stage for this task?

Investigating the statistics within the dataset takes place within the analysis phase.

8. **Visualization building:** Analyze ▾ and Construct ▾

Why did you select these stages for this task?

Analysing the data to determine important variables. Then, visualise the variables to make it easy for non technical members of the team to understand valuable insights.

9. **Write a project proposal:** Plan ▾

Why did you select this stage for this task?



Writing the project proposal is part of the planning phase and will outline the framework for the project.

10. Build a regression model: **Analyze ▾** and **Construct ▾**

Why did you select this stage for this task?

Analyze the data and determine variables. Construct a regression model to describe the relationship between variables.

11. Compile summary information about the data: **Analyze ▾**

Why did you select this stage for this task?

Summary information about the data will be communicated to team leads and management overseeing the project.